Matrix inequalities and norms

Thompson-type domination for the arithmetic symmetric modulus

Difficulty: challenging **Importance:** interesting to specialist

Problem statement

For $X \in \mathbb{C}^{n\times n}$, define $|X| = (X^*X)^{1/2}$ and

$$S(X) = \frac{|X| + |X^*|}{2}.$$

For every $n \geq 1$ and every $X, Y \in \mathbb{C}^{n\times n}$, do there exist unitary $U, V \in \mathbb{C}^{n\times n}$ such that

$$S(X+Y) \leq \sqrt{2}(US(X)U^* + VS(Y)V^*)?$$

Here $P \leq Q$ means $Q - P$ is positive semidefinite. The conjectured universal factor $\sqrt{2}$ is already known to be necessary.

Why it matters

Symmetrizing the left and right polar factors removes their directional preference. The question asks for a precise matrix-order bound for sums, which would yield singular-value and norm consequences for these positive representatives.

References

1. T. Zhang, *Operator symmetric moduli and sharp triangle inequalities*, arXiv:2603.01046v1 (1 March 2026), §1.2, Conjecture 1.6. Primary text.
2. J.-C. Bourin and E.-Y. Lee, *Some hybrid matrix triangle inequalities*, arXiv:2606.29188v1 (28 June 2026), introduction, Theorems 1.1 and 1.3. Primary text.

Status check — 2026-09-08

Both latest arXiv records remain v1. The June paper supplies related estimates but no proof of the displayed two-unitary inequality. Searches included *Zhang Conjecture 1.6 symmetric moduli*, *arithmetic symmetric modulus Thompson conjecture 2026*, and the two titles with *proof*. No resolution was located. The analogous theorem for the quadratic symmetric modulus is proved and must not be confused with this conjecture; a formula labeled Conjecture 3.13 in arXiv:2606.15624 appears to misidentify the modulus, so the draft follows Zhang's original statement.